REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Convex and Concave Lenses Practice Questions

Grade 10 Advanced | Module 16 Lesson 4 | Convex and Concave Lenses

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: 1/f = 1/do + 1/di; m = hi/ho = −di/do. Convex f positive; concave f negative. Real images have positive di.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 26–35.

P01. A convex lens in air is typically thicker:

A. At its centre than at its edges B. At its edges than at its centre
C. Only on one side of its frame D. By exactly one wavelength everywhere

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P02. A concave lens in air usually makes parallel rays:

A. Converge to a real focus B. Diverge
C. Stop completely D. Reflect backward along all paths

P03. A real object viewed through a single concave lens has an image that is:

A. Real inverted and enlarged B. Real upright and equal-sized
C. Virtual upright and diminished D. Virtual inverted and enlarged

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P04. Nearsightedness is commonly corrected with a:

A. Convex lens B. Plane mirror
C. Prism alone D. Concave lens

P05. Chromatic aberration results mainly from:

A. Different wavelengths having different focal lengths B. Light having no frequency
C. A lens having zero thickness everywhere D. Equal refraction for all colours

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P06. A thin convex lens has f = +10 cm and do = +30 cm. Calculate its signed image distance.

A. 15 cm B. 30 cm
C. 7.5 cm D. -15 cm

P07. A thin concave lens has f = -10 cm and do = +30 cm. Calculate its signed image distance.

A. -15 cm B. -7.5 cm
C. -3.75 cm D. 7.5 cm

P08. A lens has object distance +24 cm and real image distance +12 cm. Calculate its signed magnification.

A. -1 B. -0.25
C. -0.5 D. 0.5

P09. A lens produces magnification −2.0 for an object 4 cm tall. Find its signed image height.

A. -16 cm B. -4 cm
C. 8 cm D. -8 cm

P10. A lens forms a real image at 30 cm for an object 60 cm away. Calculate its focal length.

A. 20 cm B. 40 cm
C. 10 cm D. -20 cm

P11. Define a lens.

P12. Distinguish a real image from a virtual lens image.

P13. How is farsightedness commonly corrected?

P14. Describe spherical aberration.

P15. Why do compound achromatic lenses reduce chromatic aberration?

P16. A thin convex lens has f = +15 cm and do = +45 cm. Calculate its signed image distance.

P17. A thin convex lens has f = +20 cm and do = +60 cm. Calculate its signed image distance.

P18. A thin concave lens has f = -15 cm and do = +45 cm. Calculate its signed image distance.

P19. A thin concave lens has f = -20 cm and do = +60 cm. Calculate its signed image distance.

P20. A lens has object distance +36 cm and real image distance +18 cm. Calculate its signed magnification.

P21. A lens has object distance +48 cm and real image distance +24 cm. Calculate its signed magnification.

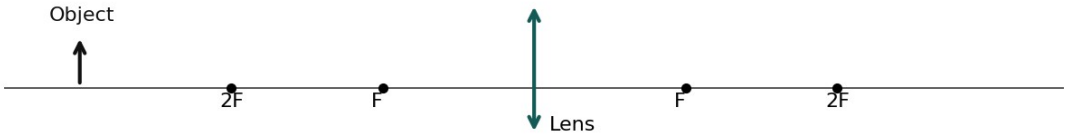
P22. A lens produces magnification −2.0 for an object 6 cm tall. Find its signed image height.

P23. A lens produces magnification −2.0 for an object 8 cm tall. Find its signed image height.

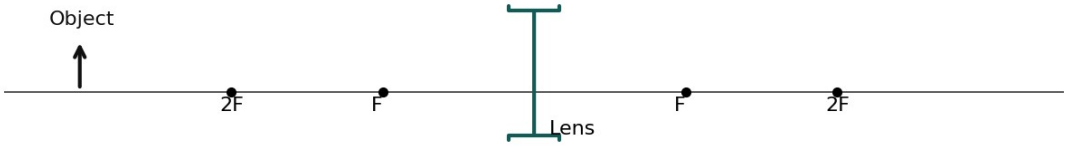
P24. A lens forms a real image at 45 cm for an object 90 cm away. Calculate its focal length.

P25. A lens forms a real image at 60 cm for an object 120 cm away. Calculate its focal length.

P26. Draw a ray diagram for an object beyond twice the focal length of a convex lens.



P27. Draw the image of a real object through a concave lens and describe its properties.



P28. Why can a convex lens serve as a magnifying glass when the object lies inside F?

P29. A student says a negative image distance always means the lens is concave. Evaluate the claim.

P30. Explain why a camera normally uses a real image rather than a virtual image at its sensor.